

Problem Definition: A small local shop uses a spreadsheet to track product details and stock levels. During busy sales periods, manually adding item prices and calculating discounts slows transactions and increases the risk of errors. This program provides shop staff with a simple, command-line tool that reads product data from their spreadsheet, adds item prices, applies a discount for purchases of £100 or more, and clearly displays the final total to the user.

Title: Small Shop Helper Program Planning & Design

1. Inputs & Outputs

Input: Users interact with the program by entering the product ID and through menu selection:

- Press 'p' to view the product list
- Press 'i' to view the inventory
- Press 'ENTER' to add a new item
- Press 'q' to exit the program

Output (What the user sees):

- A main menu displayed in the terminal
- A list of product IDs and product descriptions
- A list of product IDs, descriptions, and stock amounts
- Error messages for invalid menu choices
- The total of the transaction in message when exiting the program

2. Program Structure

The following structure allows the program to continuously run until the user chooses to exit:

1. Import required modules (csv, os)
2. Define variables and file paths
3. Load product data from a CSV file into a list
4. Display a menu inside a loop
5. Respond to user input using selection statements
6. Apply discount to append total
7. Exit the program when the user selects 'q'

3. Algorithms

Start program

Load product data

- Get the folder location of the program
- Open products.csv
- Read each row from the file
- Store all products in a list

Set up empty values

- Create an empty cart list
- Set total to 0

Repeat until user chooses to quit

- Display main menu:
 - Press 'p' to view product list
 - Press 'i' to view inventory
 - Press 'ENTER' to add an item
 - Press 'q' to exit
- Ask user to select an option

If user presses 'p'

- Display all Product IDs and Product Descriptions

Else if user presses 'i'

- Display Product ID, Product Description, and Amount in Stock

Else if user presses 'ENTER'

- Ask user to enter a Product ID
- Search through the product list
- If Product ID is found:
 - Get the product price
 - Add (product name, price) to the cart
 - Add price to total
 - Display confirmation message
- Else:
 - Display "Invalid Product ID" message

Else if user presses 'q'

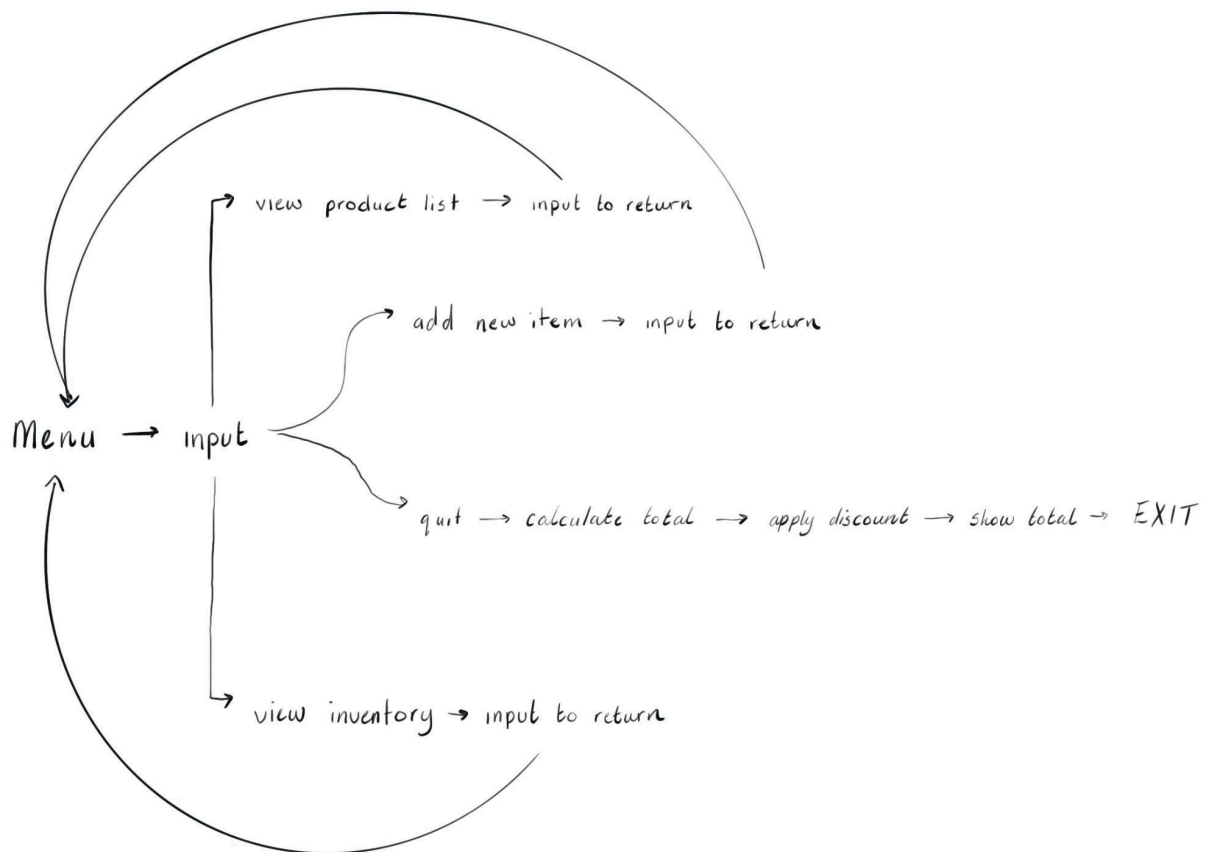
- Display invoice header
- Display each item and its price
- Display subtotal
- If subtotal is greater than or equal to £30:

- Calculate 10% discount
- Subtract discount from total
- Display discount amount
- Display final total
- Display date and time
- Display thank-you message
- End loop

Else

- Display “Invalid option” message

End program



4. Screen Layout

- **Main Menu Layout:**

--- MAIN MENU ---

Press 'p' to view the product list

Press 'i' to view the inventory

Press ENTER to add a new item

Press 'q' to exit the program

- **Product List:**

--- PRODUCT LIST ---

- 1 Face Cream
- 2 Shampoo
- 3 Conditioner
- 4 Body Scrub
- 5 Lip Balm
- 6 Facial Cleanser
- 7 Serum
- 8 Face Mask

- **Inventory:**

--- INVENTORY ---

- 1 Face Cream 45
- 2 Shampoo 120
- 3 Conditioner 95
- 4 Body Scrub 60
- 5 Lip Balm 200
- 6 Facial Cleanser 80
- 7 Serum 40
- 8 Face Mask 150

5. Data Handling

Variables:

- my_list: stores all product data from the CSV file
- abs_dir: stores the absolute directory of the program
- product_file: stores the CSV file name
- choice: stores the user's menu selection

Data Storage:

- Product data is stored in a CSV file (products.csv)

6. Control Structures

Selection (if / else):

- Used to determine which menu option the user selects
- Controls whether product lists, inventory, or exit messages are displayed
- Applied when total is equal to or greater than 100

Iteration (loops):

- A 'while True' loop keeps the menu running until the user chooses to exit

7. Error handling

Potential Error 1: If the user enters an invalid menu option, the program will display:

“Invalid option. Please try again.”

Potential Error 2: If the user enters an invalid ID when adding an item the program will display:

“Invalid product id. Please check the product listing for further details.”

KEY NOTE: BEFORE PASSING THIS PROGRAM ONTO THE USER/OWNER THEY WILL BE GUIDED ON HOW TO USE IT AND ADVISED NOT TO DELETE OR RENAME THEIR SPREADSHEET.

8. Conclusion

This design outlines a simple and practical Small Shop Helper program that reads product data from a CSV file. There will be no graphics with the user only interacting with the terminal in a Python file. Utilising basic python programming such as variables and loops.